

GARV ARORA

Robotics & Embedded Systems Engineer | Autonomous Systems | ADAS

Vellore, India | +91 88008-12254 | garvarora0205@gmail.com | linkedin.com/in/gaminization | github.com/gaminization

SUMMARY

Computer Science undergraduate (IoT) at VIT promoted to R&D Lead at SEDS VIT — the apex body of SEDS India — after 2+ years building ROS2 autonomous navigation systems, CUDA-accelerated perception pipelines, and ARM bare-metal firmware as an autonomous systems developer. Ranked 12th (IRC 2025) and 16th (IRC 2026) globally among 100+ international teams. Targeting roles in robotics software, embedded systems, ADAS, or autonomous vehicle engineering.

EDUCATION

Vellore Institute of Technology | Vellore, India

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT)

CGPA: 8.37 / 10.0

August 2023 – July 2027

Gems International School | Gurugram, India

Central Board of Secondary Education (CBSE)

Class 10: 89.3% | Class 12: 76.4%

November 2016 – April 2023

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, Bash, Verilog, C#, HTML/CSS

Robotics & Autonomous Systems: ROS2, Nav2, MoveIt2, RTAB-Map, Micro-ROS, SLAM, EKF/UKF, PID Control, Point Cloud Processing, mmWave Radar

Perception & ML: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, HuggingFace Transformers

Embedded & Infrastructure: ARM Cortex-M, ESP32, I2C/SPI/UART, FreeRTOS, Docker, Git, GitLab CI/CD, Linux, AWS, PyQt, MongoDB

WORK EXPERIENCE

SEDS Projects VIT | SEDS India

April 2024 – Present

Research & Development Lead (Promoted from Developer)

January 2026 – Present

- Promoted to R&D Lead directing 2 competition teams (50+ engineers) — Team Ardra (ISDC) and Team Vyadh (IRC/IRDC) — at SEDS VIT.
- Guided 3 sub-team leads through weekly design reviews and hardware debugging sessions, resolving 10+ critical system issues ahead of IRC, IRDC, and ISDC events.

Autonomous Systems Developer

April 2024 – January 2026

- Architected a full ROS2 navigation stack (Nav2, RTAB-Map Visual SLAM, RealSense D455) generating real-time 3D elevation maps and cutting manual teleoperation time by 70%.
- Slashed CV pipeline latency by ~60% and raised throughput 4x via CUDA, YOLOv8, ArUco/AprilTag detection, and Linux V4L2 tuning across 4 camera streams.
- Flashed Micro-ROS firmware to ESP32 units for distributed sensor acquisition and actuator control; fused IMU and encoder data with an EKF, cutting odometry drift by ~40% and reaching sub-10 cm position accuracy.
- Shipped a PyQt5 mission dashboard featuring live telemetry plots, parameter tuning controls, and FSM-based task sequencing for autonomous mission runs.

Wissen Baum Engineering Solutions

May 2025 – July 2025

Software Automation Intern

Pune, India

- Authored a BDD test suite (Python, Behave, pytest) covering 40+ scenarios, raising test coverage from 0% to 100% across 3 product flows in 6 weeks.
- Parallelized GitLab CI/CD across 3 stages (Playwright + Cypress), cutting run time by ~35% (18 to 12 min); automated QA validation with pandas/NumPy across 10,000+ rows per run.

PROJECTS

mmWave Earthquake Survivor Detection Robot | ROS2, Python, Hi-Link LD2410C, ESP32

2025

- Integrated an LD2410C mmWave radar with ROS2 to detect survivor breathing through rubble, fusing radar data with IMU odometry to mark locations with <15 cm accuracy.

Gesture-Controlled 5-DOF Robotic Arm | C++, Arduino, MPU6050, Flex Sensors

2024

- Mapped MPU6050 orientation and 2 flex sensor readings to 5-DOF servo commands with <50 ms latency, achieving 94% gesture accuracy across 8 poses on 200+ samples.

HayaiOS – Bare-Metal RTOS | C, ARM Assembly, ARM Cortex-M4

2026

- Implemented a preemptive scheduler (1 ms tick), context switching in under 50 assembly instructions, mutex/semaphore primitives, and a HAL for all register-level I/O.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

2025

- Stabilized KinectFusion TSDF tracking by fusing RealSense D455 depth frames with IMU via complementary filter, cutting tracking failures by ~50% on a 6-DOF fast-motion path.

CERTIFICATIONS & ACHIEVEMENTS

IRC 2025: 12th globally (100+ teams) | **IRC 2026:** 16th globally | **IRDC 2025:** Top 5 Finalist | **Oracle Cloud 2025:** Generative AI Professional | Data Science Professional